

Question C.3 — Invariance of the Evidence Budget

All numerical measurements in this section come from the instrumented campaign of 2000 runs (20 amplitudes $\times$ 100 seeds, $H = 2000$, $M = 10$ trees in the adaptive random forest, ARF, baseline $\hat{p}_0$ estimated over the last 3000 pre-drift steps) documented in `04_experiments/resultats_R2/`. Every number quoted is reproduced by `scripts/chiffres_QC.py` from the tables in `resultats/data/`. The deterministic hierarchy $\tau_{\text{ARF}} \leq \tau_{\text{swap}}(q)$ (Q C.1) and the dimensional analysis with its detection certificate (Q C.2) are assumed established. τ^* denotes the drift instant, from which every duration is counted; τ_{rec} denotes the return of the seed-averaged error below $\hat{p}_0 + \rho \Delta e$ with $\rho = 0.25$, held for 20 consecutive steps, which Q C.1 writes $\tau_{\text{err}}(\rho)$.

1 Budget Expression and Residual Exponent

The article’s stylized model assumes that after the drift the error exceeds its baseline by an amount Δe for a transient of length $W = \tau_{\text{ARF}}$, then falls back to baseline. The excess-error area offered to the detector is then $\mathbb{E}[A] = \Delta e \cdot \mathbb{E}[\tau_{\text{ARF}}]$, provided the horizon fully covers the transient, which $H = 2000$ guarantees across the whole grid: at the lowest amplitude of the grid, $\Delta e = 0.0282$, the short window of Section 4 is $w = 3 \times 18.5 / \Delta e = 1969$ steps. Plugging in the article’s empirical fit $\mathbb{E}[\tau_{\text{ARF}}] \approx 18.5 \Delta e^{-0.98}$:

$$\boxed{\mathbb{E}[A] \approx 18.5 \Delta e^{1-0.98} = 18.5 \Delta e^{0.02}} \quad (1)$$

Residual exponent and variation across the grid. The residual exponent of Δe is 0.02. Over $\Delta e \in [0.028; 0.498]$, the ratio of the extremes is $(0.498/0.028)^{0.02} = 1.059$: the predicted budget varies by 5.9%, from 17.2 to 18.2, as the shock amplitude varies by a factor of 17.7.

Formal conclusion. Amplitude and duration nearly exactly offset each other: the forest repairs itself all the faster the stronger the shock, and the product $\Delta e \times W$ stays close to constant. A violent macroeconomic shock therefore leaves the detector no more evidence than a slow drift. The budget behaves like a constant of the forest-detector pair, not as a characteristic of the observed drift.

2 The Actually Usable Budget

The detector does not consume the raw area. Its tolerance $\delta_P = 0.01$ is subtracted at every step, so the budget actually accumulable is $(\Delta e - \delta_P) W$ rather than $\Delta e W$. The confiscated share is therefore $\delta_P / \Delta e$, independent of the transient’s duration.

Δe	raw budget	share taken	usable budget
0.028 (bottom of grid)	17.23	35.5%	11.11
0.498 (top of grid)	18.24	2.0%	17.88

Where the tolerance bites. It is at the *low* end of the grid that δ_P takes a non-negligible share: more than a third of the budget at $\Delta e = 0.028$, versus 2% at the other end. Since the tolerance is an absolute threshold, it weighs all the more heavily the weaker the signal. It therefore *worsens* the invariance rather than correcting it: the usable budget is no longer constant, it decreases toward low amplitudes, exactly where the detector would need it most. Figure 1 places the two predicted budgets against the measured short-window area across the grid.

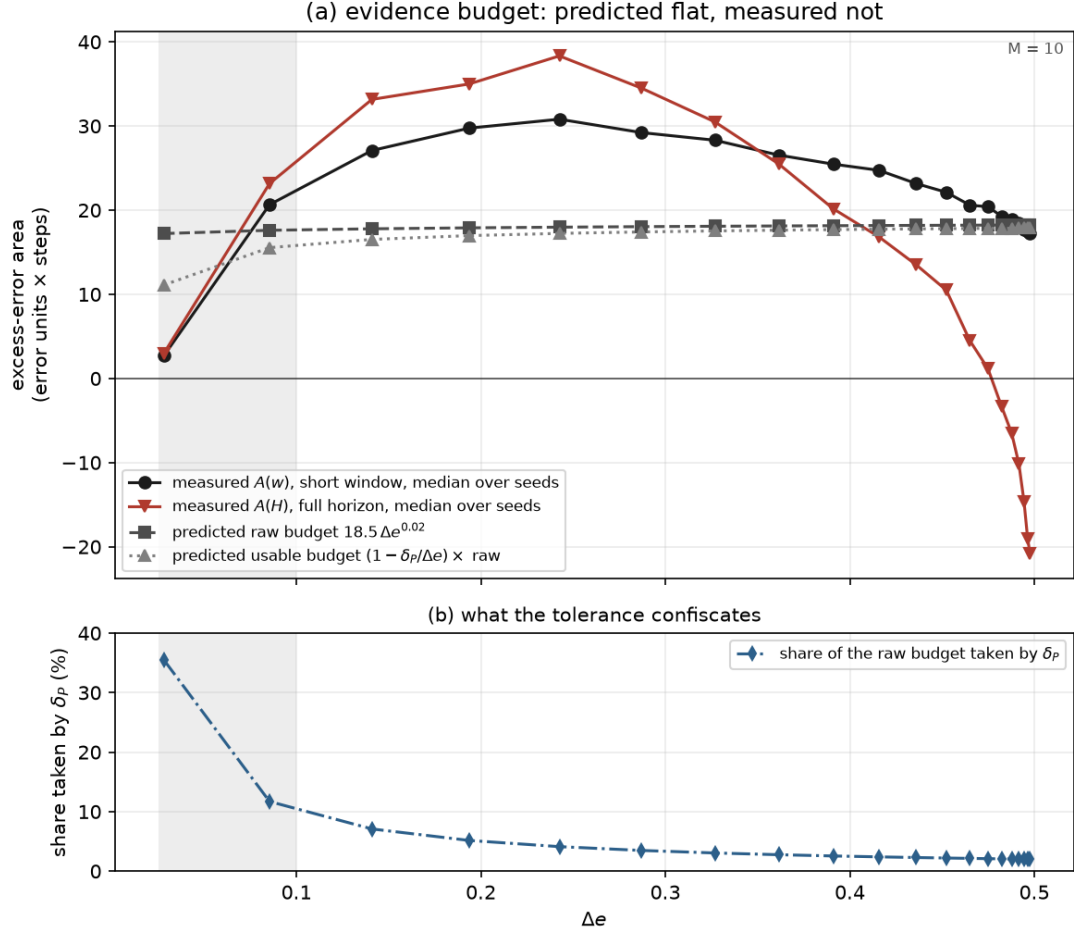


Figure 1: (a) The two predicted budgets, raw $18.5 \Delta e^{0.02}$ and usable $(1 - \delta_P/\Delta e)$ times the raw one, against the two measured areas, over the short window $A(w)$ and over the full horizon $A(H)$, medians over 100 seeds. (b) The share of the raw budget confiscated by the tolerance δ_P . What to see in (a): the two predicted curves are flat, neither measured one is, $A(w)$ peaks near $\Delta e = 0.243$ and $A(H)$ crosses zero from $\Delta e = 0.482$; in (b): the tolerance takes more than a third of the budget at the bottom of the grid and under 3% over most of it. Series are distinguished by marker and line style, not by colour. The shaded band marks the two amplitudes outside the decision domain of Q D.4. Table QCD_budget_preuve_full.

3 Confrontation with the Full-Window Certificate

The certificate established in Q C.2 requires, over the full window, a raw area of at least $\lambda + H\delta_P$. With $H = 2000$ and $\delta_P = 0.01$ the toll is 20 regardless of the threshold, giving 28, 45 and 70 for $\lambda \in \{8, 25, 50\}$, to be compared to the budget of about 18 established in (1). **None of the three thresholds is reachable**, the lowest already requiring one and a half times the available budget. The required evidence does not exist in the signal, whatever the setting chosen.

What happens to the certificate as the horizon lengthens. The required threshold $\lambda + H\delta_P$ grows *linearly* in H , while the budget stays *constant* per (1). The gap therefore widens indefinitely: once the error has recovered, every additional observation step adds δ_P to the toll without contributing anything to the numerator, so that observing longer degrades the certificate instead of improving it. Hence the value of anchoring it on a short window $w = 3 \times 18.5/\Delta e$, sized to the transient of the amplitude under consideration.

4 Verification Against the Measurements

Panel (a) of Figure 1 carries the comparison this section needs: the measured areas $A(w)$ and $A(H)$ against the constant predicted by (1), on the full grid.

The invariance holds in sign, not in value. Over $\Delta e \in [0.085; 0.498]$, the shock amplitude varies by a factor of 5.8, and absent forest adaptation the area would vary by as much. It actually varies by a factor of **1.78**: the compensating effect is indeed present. But the model predicts a factor of 1.04, and it underestimates the budget in the middle of the grid, with 30.8 measured against 18.0 predicted at $\Delta e = 0.243$. The invariance is thus confirmed in principle and approximate in its values.

Why the lowest amplitude is set aside, and what it costs. The factor of 1.78 is computed over $[0.085; 0.498]$, not over the full grid. Including $\Delta e = 0.028$, where $A(w)$ collapses to 2.71, the measured area varies by a factor of **11.4** instead, and the invariance no longer holds in any useful sense. The exclusion is not a matter of convenience: at that amplitude the short window $w = 1969$ steps nearly covers the whole horizon, so $A(w)$ and $A(H)$ stop being distinguishable and the short-window construction loses its meaning. But the restriction must be stated with the result, since it is what separates a factor of 1.8 from a factor of 11.

The full-window area turns negative. $A(H)$ peaks at 38.3 then decreases, crosses zero from $\Delta e = 0.482$ onward and drops to -20.8 . The explanation lies in the geometry of the problem rather than in the measurement. At large amplitude the post-drift boundary is $x_0 + x_1 > 4$ with $x_0 + x_1 \sim \mathcal{N}(0, 2)$: class 1 becomes nearly absent and the post-drift task turns out to be easier than the original one. Over the last 50 steps of the horizon, the seed-averaged error is 0.0084 at $\Delta e = 0.482$, 0.0072 at 0.488, 0.0078 at 0.492, then 0.0060, 0.0040 and 0.0028 at 0.494, 0.496 and 0.498, against a pre-drift baseline of 0.0231; the decrease is not monotone in Δe , and the values over the last 100 or 500 steps are within 0.001 of these (table QCD_fin_horizon_full). What produces the deficit is not the depth of that undershoot but its duration: at $\Delta e = 0.498$ the seed-averaged error is below the baseline on 1934 of the 2000 steps, and it stops rising above it at all from step 1641 onward; at $\Delta e = 0.452$, the lowest amplitude of this band, it is below on 1459 steps. The deficit $A(H) - A(w) = -20.83 - 17.31 = -38.15$ (table QCD_budget_preuve_full) is what reconciles a short-window area of 17.3 with a full-horizon area of -20.8 ; it grows monotonically across the band, from -11.63 at $\Delta e = 0.452$ to -38.15 at the top. The stylized model is therefore at fault at large amplitude, where the error drops below its initial baseline instead of returning to it. This is a second reason, independent of the one given in Q C.2, to certify only on the short window.

Where does the gap come from: the baseline, the censoring, or the fit? The subject invites us to look first at $\hat{p}_0$. Its estimation error propagates in a perfectly correlated way across the whole area, with a bias $-H(\hat{p}_0 - p_0)$; estimated over 3000 steps, $\hat{p}_0$ has an inter-seed standard deviation of 0.00298, i.e. ± 6.0 on $A(H)$. This is non-negligible against a budget of 18, but insufficient to account

for the gap of 14.5 observed at $\Delta e = 0.028$, especially since this bias has a random sign while the observed gap is systematic.

The test that truly settles it is to compare the measured τ_{ARF} to its own prediction, since (1) stacks two distinct hypotheses: the model $\mathbb{E}[A] = \Delta e \mathbb{E}[\tau_{\text{ARF}}]$ on one hand, and the power-law fit on the other.

Δe	measured τ_{ARF}	predicted	ratio	$\Delta e \times \tau_{\text{ARF}}$	measured $A(w)$
0.028	118	611	0.19	3.33	2.71
0.141	263	126	2.08	37.07	27.09
0.287	64	63	1.02	18.37	29.22
0.436	34	42	0.81	14.82	23.16
0.498	28	37	0.76	13.93	17.31

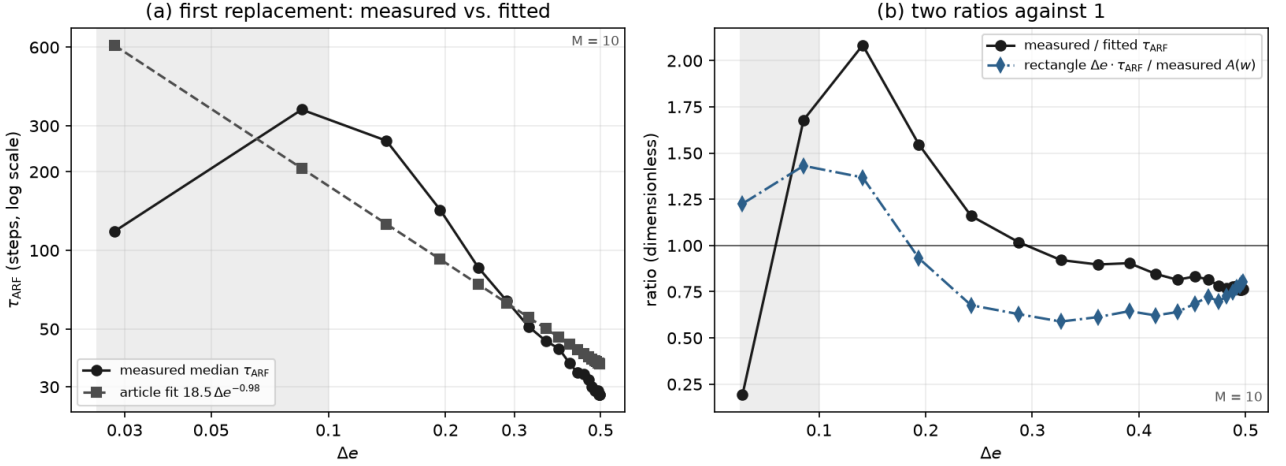


Figure 2: The two stacked hypotheses of (1), on all 20 amplitudes rather than the five rows of the table. (a) Measured median τ_{ARF} against the article’s fit $18.5 \Delta e^{-0.98}$, log-log axes: the two cross only near $\Delta e \approx 0.29$, and the measured curve rises before it falls. (b) The ratio measured over fitted τ_{ARF} , and the ratio of the rectangle $\Delta e \times \tau_{\text{ARF}}$ to the measured $A(w)$, against the line at 1: the rectangle crosses 1 between $\Delta e = 0.141$ and 0.194 , which is the sign reversal discussed below. Table QCD_diagnostic_ajustement_full.

At the lowest amplitude, the fit predicts the first replacement at step 611 whereas it occurs at step 118, five times earlier. The power-law fit is therefore at fault, and the gap is not localized: the measured-over-predicted ratio is 0.19 at the bottom of the grid, exceeds 2 around $\Delta e = 0.141$, only matches around $\Delta e = 0.287$, then falls back to 0.76 at the top.

But the underlying model fails too. It would be convenient to conclude that only the fit is at fault and that the model $\mathbb{E}[A] = \Delta e \mathbb{E}[\tau_{\text{ARF}}]$ survives. The measurements do not support this. Recomputing $\Delta e \times \tau_{\text{ARF}}$ from the *measured* τ_{ARF} at every amplitude and comparing it to the measured $A(w)$, the model overestimates the area by 23% to 43% on the three amplitudes below $\Delta e = 0.19$, and underestimates it on the seventeen amplitudes above: by 6.9% at $\Delta e = 0.194$, the first amplitude past the crossing, then by 19.5% to 41.1% from $\Delta e = 0.243$ onward. It changes sign between $\Delta e = 0.141$ and 0.194 and coincides with the measurement nowhere on the grid (Figure 2b). At $\Delta e = 0.028$ it gives 3.33 against 2.71 measured, and at $\Delta e = 0.287$, 18.4 against 29.2.

The reason is the one Q C.1 establishes: $W = \tau_{\text{ARF}}$ is not the duration of the transient. τ_{ARF} dates the *first* tree replacement, whereas three-quarters of the forest is renewed 6.0 to 14.3 times later. Taking the first replacement as the width of the error plateau is inconsistent with the hierarchy proved in Q C.1, and the sign reversal at $\Delta e \approx 0.19$ is the visible trace of that inconsistency. Both hypotheses stacked in (1) therefore fail, each in its own range.

Consequence for the report. The budget’s invariance rests on two successive hypotheses, and *neither* survives the measurements. The model $\mathbb{E}[A] = \Delta e \cdot \mathbb{E}[\tau_{\text{ARF}}]$ misses the measured area by -41% to $+43\%$ and reverses sign around $\Delta e = 0.19$, because τ_{ARF} is not the width of the transient. The power-law fit $18.5 \Delta e^{-0.98}$ only holds near the middle of the grid. The conclusion “a violent shock leaves no more evidence than a slow drift” is nonetheless *confirmed*, but by direct measurement alone: the measured area varies by a factor of 1.8 where the amplitude varies by 5.8. It can no longer be derived from the 0.02 exponent, nor from the stylized rectangle. Both are scaffolding that the measurement has to replace rather than support.

The theoretical baseline. The pre-drift target $y = \mathbf{1}\{x_0 + x_1 > 0\}$ is a *deterministic* function of the inputs, with no label noise: the Bayes error is exactly zero. The measured baseline, 0.0231, is therefore entirely approximation error: Hoeffding trees cutting an oblique boundary with axis-perpendicular splits. The detector thus monitors deviations from a reference level that depends on the classifier itself: a better-fitted forest would lower the baseline and change the evidence budget with no property of the drift having changed.

5 What This Section Establishes

The evidence budget offered to the detector is nearly independent of the drift’s amplitude: the area varies by a factor of 1.8 as the amplitude varies by a factor of 5.8, where it would track the latter absent adaptation. The tolerance δ_P amputates this budget at low amplitudes, and lengthening the observation horizon widens the deficit further.

Three caveats accompany this result. The power-law fit on which the article bases the invariance is only valid near the middle of the grid, so the argument must rest on direct measurement rather than on the 0.02 exponent. The stylized rectangle $A = \Delta e \times \tau_{\text{ARF}}$ is itself wrong, by -41% to $+43\%$ with a sign reversal, because τ_{ARF} dates the first replacement and not the end of the transient. And the model is at fault at large amplitude for a third, independent reason: the error drops below its initial baseline instead of returning to it. The invariance reported here is an empirical fact about the measured area, established directly and stated over $\Delta e \in [0.085 ; 0.498]$; it is not a consequence of the article’s derivation.